

VIGNESH RAM SIVAKUMAR

London, UK | +44 7818460941

✉ vigneshsiva9889@gmail.com

in [linkedin.com/in/vignesh-ram-sivakumar](https://www.linkedin.com/in/vignesh-ram-sivakumar)

🐙 github.com/V1cky16112003



vigneshramsivakumar.me

Projects

Multimodal Emotion Recognition on SEED-VII: Reproducibility Critique & Honest Baseline | *MSc Dissertation, KCL* / *Python, PyTorch, Graph Neural Networks, Self-supervised Learning, SLURM/HPC, EEG Signal Processing, Statistical Hypothesis Testing* 2025-2026

- Audited a published IEEE TAFFC 2025 benchmark (71.28% claimed accuracy) by replicating the architecture on King's College London's A100 GPU cluster; proved the reported figure is an evaluation-leakage artefact unreachable under any leak-free protocol
- Designed a partial-leakage block-size sweep across 20 subjects, tracing a quantified leakage-accuracy curve that reproduced the original figure to within 0.17 pp at block size approx 44; confirmed via paired Wilcoxon tests ($W=0$, $p<2e-6$, Cohen's $d=3.72$)
- Established honest baselines: $53.40\% \pm 9.77\%$ (paper-faithful protocol) and $48.20\% \pm 0.90\%$ (fully train/test-isolated, 3-seed average); oracle hyperparameter ceiling of 58.21% still 13 pp short of the claimed figure
- Validated the leakage mechanism on an independent dataset (MultiPhysio-HRC, 44 participants, industrial EEG): +13.81 pp inflation ($p=1.15e-8$, $d=1.115$); honest result indistinguishable from majority-class floor
- Benchmarked three mechanistically independent methods under the honest protocol: graph-convolutional model (DAMS-GCT: 42.34%), masked self-supervised pretraining (42.48%), and intensity-aware multitask MAET (47-48%); convergent nulls confirm the honest ceiling is feature-driven, not architecture-limited
- Delivered a full reproducibility package: pinned dependencies, fixed seeds, canonical evaluation path, offline self-test, and unit tests for multitask training paths

ArXiv Research Intelligence Agent (Production RAG System) | *Personal Project* 2026
/ *Python, FastAPI, LangGraph, PostgreSQL (pgvector), sentence-transformers, Groq, Gemini 2.5 Flash, Upstash Redis, React 18, Vite, Docker, GitHub Actions, MLflow*

- Engineered a four-node LangGraph agent (Planner -> Executor -> Critic -> Reporter) with a Critic-driven retry loop up to 3 Executor re-runs before forcing a final grounded answer with citations
- Shipped a full-stack platform over 50,000 ArXiv ML papers: FastAPI backend on Hugging Face Spaces, React 18/Vite frontend on Vercel, with GitHub Actions CI/CD running lint, tests, and a RAGAS quality gate on every push
- Built the RAGAS evaluation pipeline with a 20-question golden set — the CI gate blocks merges when faithfulness drops below 0.80 or answer relevancy below 0.75 — with metrics logged to MLflow on DagsHub
- Designed a two-stage ingestion pipeline loading JSONL metadata into Neon Postgres, chunking abstracts and embedding with Nomic-embed-text-v2 (768-dim) into pgvector with HNSW indexing
- Built an LLM routing layer — Groq (Llama 3.3 70B) as primary with Gemini 2.5 Flash automatic fallback on rate limits — plus Upstash Redis caching for response deduplication

Conversational Multimodal Image Recognition Chatbot | *SRM Institute of Sci*-Jan 2025 - May 2025
ence and Technology / *Gemini 2.0 Flash, Azure Cognitive Services, FastAPI, React.js, Python, OCR, NLP, TTS, STT*

- Built a multimodal chatbot combining real-time image recognition (Gemini 2.0 Flash), OCR (Azure Computer Vision), and voice I/O (Azure Speech) with support for English, Tamil, Hindi, and Telugu
- Designed a production REST API (FastAPI backend, React.js frontend) that handles real-time multimodal inference across 4 languages with sub-second response latency

Technical Skills

Programming: Python (proficient), SQL, JavaScript, Git

Agentic AI & LLM Systems: LangGraph agentic workflows, RAG (pgvector, RAGAS), LLM routing (Groq, Gemini), FastAPI deployment

AI Security: OWASP Agentic/LLM Top 10 mapping, Cedar policy, adversarial stress-testing, information-flow modelling

ML & AI: PyTorch, TensorFlow, Scikit-Learn, GNNs, Transformers, HuggingFace, NLP, Computer Vision, Multimodal AI

Engineering & Backend: FastAPI, Docker, REST APIs, PostgreSQL, CI/CD (GitHub Actions), MLflow, Azure Cognitive Services, MySQL

Signals & Modelling: EEG/EDA/BVP signal processing, Multimodal fusion, Uncertainty quantification, Transfer learning

Frontend & Viz: React.js, Matplotlib, Seaborn

Work Experience

Guard AI	Jun 2026 - Jul 2026
<i>AI Security Researcher / Policy Architect</i>	<i>Stealth Startup</i>
- Designed and stress-tested the authorization policy layer of a runtime security system mediating every action an autonomous AI agent proposes — returning one of five decisions (allow, deny, transform, sandbox, escalate) before any action reaches the real world - Authored a complete Cedar-ready policy specification mapping the full OWASP Agentic Top 10 (ASI01-ASI10) and OWASP LLM Top 10 threat classes onto an enumerated action x resource decision matrix with per-severity asymmetric acceptance thresholds - Architected a core information-flow model — structural provenance tracking (trust from source channel, never from content), a two-axis authorization framework separating recipient authorization from boundary-crossing, and a bounded declassification lattice — validated against six real production incidents including EchoLeak (CVE-2025-32711, CVSS 9.3), MCP Tool Poisoning, and Cursor Studio confused-deputy (CVE-2026-21520) - Ran three adversarial stress-test passes on the specification, resolving 20+ internal contradictions and security gaps — including a data-launders path through the declassification layer, a stale-authorization race condition at the Cedar decide/execute boundary, and a null-sentinel collision in recipient-authorization logic - Specified fail-closed runtime computation contracts for seven load-bearing evidence-layer inputs ensuring correct policy decisions cannot be silently defeated by mis-computed inputs	

4i Apps Solutions Private Limited	Dec 2023 - Jan 2024
<i>AI Intern (Clustering and Deep Learning Analysis)</i>	<i>Chennai, India</i>
- Built a K-Means segmentation pipeline on ~3-4 GB of purchase data, surfacing customer behavioural patterns and translating findings into targeting recommendations adopted by the business team - Trained and evaluated TensorFlow/Keras deep learning models for image classification and NLP, owning each model end-to-end from raw data through preprocessing, training, and final performance reporting	

4i Apps Solutions Private Limited	Jul 2023
<i>Data Science Intern (Data Science & Machine Learning)</i>	<i>Chennai, India</i>
- Built an end-to-end classification pipeline on commercial datasets — data cleaning and EDA with Pandas/NumPy, then model development and train/test evaluation with Scikit-Learn - Delivered Matplotlib/Seaborn dashboards presenting model results to clients, producing the charts and summaries used in business reporting	

Publications

Survey paper on various techniques of speech recognition, transcription and summarization Supervisor: Dr. J. Selvin Paul Peter DOI: https://doi.org/10.1063/5.0331220	2024
Introduced an automated environment for lecture transcription and summarization with live transcripts, timestamps, real-time summaries, emotional intelligence across participants, translation, and multi-language support (4 local languages) using state-of-the-art Speech Recognition, Text Processing, NLP, and ML pipelines	

Certifications

Model Context Protocol: Advanced Topics Anthropic	2026
AI Fluency Framework & Foundations Anthropic	2026